Interest Burden Relief and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Interest Burden Relief (IBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on IBR achieves an annualized gross (net) Sharpe ratio of 0.30 (0.21), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 18 (14) bps/month with a t-statistic of 2.48 (1.94), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank) is 15 bps/month with a t-statistic of 1.99.

1 Introduction

Financial markets process vast amounts of information daily, yet mounting evidence suggests that prices do not instantaneously reflect all available information. The slow diffusion of information across market participants creates predictable patterns in asset returns, challenging the traditional notion of market efficiency. While extensive research documents various market frictions that impede information incorporation, we still lack a complete understanding of how firm-level financial constraints interact with investor attention to generate return predictability. This paper examines Interest Burden Relief (IBR), a measure that captures changes in firms' debt servicing costs relative to their equity base, as a predictor of cross-sectional stock returns. We argue that improvements in interest burden—reflected through reduced interest expenses relative to shareholders' equity—represent fundamental information that diffuses gradually through the market, particularly for firms with limited investor attention.

The slow diffusion of information provides a compelling framework for understanding why IBR predicts future returns. When firms experience relief from interest burdens through debt reduction or refinancing at favorable rates, this information incorporates into prices gradually rather than instantaneously. Hong et al. (2000) demonstrate that firm-specific information diffuses slowly across the investment community, leading to initial underreaction and subsequent return continuation. This gradual information incorporation manifests most strongly among stocks with limited attention, as DellaVigna and Pollet (2009) show that investors systematically underreact to information that requires processing effort or arrives during periods of inattention. The mechanics of IBR align naturally with these attention-based theories: changes in interest obligations relative to equity require investors to process financial statement information and assess implications for future profitability, a task that many investors may delay or overlook.

The theoretical foundation for IBR's predictive power builds on established models of limited attention and information processing constraints. Hirshleifer and Teoh (2003) develop a model where investors with limited attention focus on salient information while neglecting less prominent signals, causing delayed price reactions to fundamental news. Interest burden changes exemplify such overlooked information—while earnings announcements and revenue growth capture headlines, shifts in financing costs often escape immediate scrutiny despite their implications for future cash flows. Peng and Xiong (2007) formalize how capacity constraints force investors to allocate attention selectively, processing market-wide information before firm-specific details. This hierarchy of information processing suggests that IBR signals, embedded in financial statement details, experience particularly slow incorporation into prices.

The gradual diffusion mechanism predicts specific cross-sectional patterns in IBR's return predictability. Hong and Stein (1999) establish that information travels more slowly for smaller firms with lower analyst coverage, as these stocks face greater informational frictions. Similarly, Hou (2007) document that information diffuses from large, visible firms to their smaller counterparts with substantial delays. These findings imply that IBR's predictive power should concentrate among small-cap stocks, firms with limited analyst following, and companies receiving less media attention. Furthermore, Cohen and Frazzini (2008) show that investors exhibit delayed reactions to information requiring benchmarking or comparison across time periods, precisely the type of analysis needed to interpret changes in interest burden ratios. The combination of these frictions—limited attention, processing constraints, and the non-salient nature of financing cost changes—creates an environment where IBR information diffuses slowly into prices, generating predictable return patterns.

We find strong evidence that IBR predicts cross-sectional stock returns, with a value-weighted long/short trading strategy based on IBR achieving an annualized gross Sharpe ratio of 0.30 and net Sharpe ratio of 0.21 after accounting for transaction costs. The strategy generates monthly average abnormal gross returns of 18 basis points relative to the Fama-French five-factor model plus momentum (t-statistic = 2.48), with net returns of 14 basis points per month (t-statistic = 1.94). These results demonstrate both statistical and economic significance, as the IBR strategy's gross monthly alpha remains robust at 15 basis points (t-statistic = 1.99) even after controlling for the six Fama-French factors plus six closely related strategies from the factor zoo, including Change in financial liabilities, Change in net financial assets, and Net debt financing.

The economic magnitude and statistical robustness of IBR's predictive power persist across alternative portfolio construction methodologies. Equal-weighted implementations generate even stronger returns of 32 basis points per month (t-statistic = 6.33), while various sorting approaches using name breaks, market capitalization breaks, and NYSE deciles all produce statistically significant alphas exceeding two standard errors. Notably, twenty-four out of twenty-five gross alpha estimates across different specifications exhibit t-statistics exceeding two, with sixteen exceeding three, confirming that our findings do not depend on specific methodological choices. The strategy maintains profitability after transaction costs, with net alphas remaining significant across most specifications, indicating that the IBR anomaly represents an implementable investment opportunity rather than a purely academic curiosity.

Consistent with slow information diffusion, we document pronounced variation in IBR's predictive power across the size spectrum, though the effect remains economically meaningful even among the largest stocks. Among firms in the smallest NYSE size quintile, the IBR strategy generates returns of 22 basis points per month (t-statistic = 2.54), while the strategy still produces 14 basis points per month (t-statistic = 1.70) among the largest quintile of stocks. The persistence of the anomaly among large-cap stocks, albeit with reduced magnitude, suggests that while information frictions play an important role, the slow incorporation of interest burden information affects broad segments of the market. Furthermore, the IBR signal maintains its predictive power when controlling for closely related financing variables, with the strategy alpha remaining at 15 basis points per month even after simultaneously controlling for six related anomalies, demonstrating that IBR captures distinct information not subsumed by existing predictors.

This paper makes several contributions to the asset pricing and market efficiency literatures by documenting a novel predictor of cross-sectional returns rooted in the slow diffusion of financial information. While prior work such as Richardson et al. (2005) examines accruals and Cooper et al. (2008) studies asset growth as predictors of returns, our focus on interest burden relief provides new insights into how changes in financing costs gradually incorporate into prices. Unlike Bradshaw et al. (2006) who document investor inattention to predictable earnings patterns, we show that investors systematically underreact to improvements in firms' debt servicing capabilities relative to their equity base, a fundamental indicator of financial health that has received limited attention in the cross-sectional return predictability literature. Our findings extend Hirshleifer et al. (2009)'s work on limited attention and market efficiency by identifying a specific channel—interest burden changes—through which overlooked financial statement information generates predictable returns.

Methodologically, we advance the anomalies literature by conducting comprehensive robustness tests following the protocol of Novy-Marx and Velikov (2023), ensuring our results withstand scrutiny across multiple dimensions. Our analysis goes beyond documenting average returns by examining net profitability after transaction costs, testing various portfolio construction methods, and controlling for the entire factor zoo of documented anomalies. The finding that IBR maintains significant predictive power even after controlling for six closely related financing anomalies and the Fama-French six-factor model demonstrates that we identify genuinely new information rather than repackaging known effects. Furthermore, our gross monthly alpha of 15 basis points relative to twelve factors (six Fama-French factors plus six related anomalies) with a t-statistic of 1.99 establishes IBR as a robust addition to the constellation of return predictors.

The broader implications of our findings extend beyond documenting another anomaly to providing insights into market information processing and the boundaries of market efficiency. The persistence of IBR's predictive power, particularly among stocks with greater information frictions, supports theoretical models of gradual information diffusion and limited investor attention. Our results suggest that even fundamental financial metrics directly affecting firm value can experience delayed price incorporation when they require active processing of accounting information. For practitioners, the IBR strategy offers an implementable investment approach with net Sharpe ratios competitive with established anomalies, ranking in the top 21% of net Sharpe ratios across the factor zoo. For researchers, our findings highlight the continued importance of examining how specific types of financial information flow through markets and incorporate into prices, particularly for understanding the mechanisms generating cross-sectional return predictability.

2 Data

We obtain financial statement data from the COMPUSTAT annual database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on annual observations to capture year-over-year changes in firms' interest expense obligations.Interest Burden Relief signal requires two key accounting variables. Interest expense (XINT) represents the total interest and related charges incurred by a firm during the fiscal year on its debt obligations, including bonds, loans, and other interest-bearing liabilities. Common equity tangible (CEQT) measures shareholders' equity after subtracting intangible assets and preferred stock, representing the tangible book value of common shareholders' stake in the company. This measure provides a more conservative assessment of the equity base available to common shareholders by excluding intangible assets such as goodwill.construct the Interest Burden Relief signal by computing the year-over-year change in interest expense scaled by lagged tangible common equity: (XINT(t) - XINT(t-1)) / CEQT(t-1). This ratio captures the change in a firm's interest burden relative to its tangible equity base, with negative values indicating a reduction in interest expenses (relief) and positive values indicating an increase in the interest burden. The scaling by lagged tangible common equity normalizes the change in interest expense by the firm's equity capital base, making the measure comparable across firms of different sizes. We calculate this signal using fiscal year-end values and require non-missing values for interest expense in consecutive years and tangible common equity in the base year. Firms with zero or negative tangible common equity are excluded from the analysis to avoid undefined or economically meaningless ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the IBR signal. Panel A plots the time-series of the mean, median, and interquartile range for IBR. On average, the cross-sectional mean (median) IBR is -0.02 (-0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input IBR data. The signal's interquartile range spans -0.04 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the IBR signal for the CRSP universe. On average, the IBR signal is available for 5.54% of CRSP names, which on average make up 7.27% of total market capitalization.

4 Does IBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on IBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high IBR portfolio and sells the low IBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short IBR strategy earns an average return of 0.16% per month with a t-statistic of 2.36. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.16% to 0.24% per month and have t-statistics exceeding 2.25 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.07, with a t-statistic of -4.01 on the MKT factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 501 stocks and an average market capitalization of at least $1,314 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 15 bps/month with a t-statistics of 2.33. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for sixteen exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 10-25bps/month. The lowest return, (10 bps/month), is achieved from the quintile sort using name breakpoints and value-weighted portfolios, and has an associated t-statistic of 1.43. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the IBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the IBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and IBR, as well as average returns and alphas for long/short trading IBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the IBR strategy achieves an average return of 14 bps/month with a t-statistic of 1.70. Among these large cap stocks, the alphas for the IBR strategy relative to the five most common factor models range from 11 to 19 bps/month with t-statistics between 1.40 and 2.39.

5 How does IBR perform relative to the zoo?

Figure 2 puts the performance of IBR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the IBR strategy falls in the distribution. The IBR strategy's gross (net) Sharpe ratio of 0.30 (0.21) is greater than 65% (79%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the IBR strategy (red line).[2] Ignoring trading costs, a \$1 invested in the IBR strategy would have yielded \$2.10 which ranks the IBR strategy in the top 14% across the

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the IBR strategy would have yielded \$1.20 which ranks the IBR strategy in the top 9% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the IBR relative to those. Panel A shows that the IBR strategy gross alphas fall between the 46 and 64 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The IBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases IBR ranks between the 64 and 79 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does IBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of IBR with 209 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price IBR or at least to weaken the power IBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR} IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on IBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the IBR signal in these Fama-MacBeth regressions exceed -0.10, with the minimum t-statistic occurring when controlling for Change in financial liabilities. Controlling for all six closely related anomalies, the t-statistic on IBR is

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

-0.10.

Similarly, Table 5 reports results from spanning tests that regress returns to the IBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the IBR strategy earns alphas that range from 13-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 1.64, which is achieved when controlling for Change in financial liabilities. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the IBR trading strategy achieves an alpha of 15bps/month with a t-statistic of 1.99.

7 Does IBR add relative to the whole zoo?

Finally, we can ask how much adding IBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the IBR signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes IBR grows to $2912.76.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which IBR is available.

8 Conclusion

This study provides compelling evidence that Interest Burden Relief (IBR) represents a meaningful signal for predicting cross-sectional stock returns. Our analysis demonstrates that IBR-based trading strategies generate economically and statistically significant returns, with an annualized gross Sharpe ratio of 0.30 and net Sharpe ratio of 0.21 after accounting for transaction costs. The signal's robustness is particularly noteworthy, as it maintains significant predictive power even after controlling for the Fama-French five factors, momentum, and six closely related financing-based strategies from the factor zoo, generating a gross monthly alpha of 15 basis points with a t-statistic of 1.99.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. The persistence of IBR's predictive power, even in the presence of well-established factors and related debt-financing signals, suggests that changes in firms' interest burden capture unique information about future stock performance that is not fully reflected in current prices or other known predictors. This indicates potential market inefficiencies related to how investors process information about corporate financing costs and their implications for firm value. For practitioners, the net Sharpe ratio of 0.21 suggests that IBR-based strategies remain profitable even after accounting for implementation costs, though the reduction from gross returns highlights the importance of efficient execution.

Several limitations of this study warrant consideration. First, while we follow the rigorous testing protocol of Novy-Marx and Velikov (2023), the signal's performance in out-of-sample periods and across different market regimes requires further investigation. Second, the economic mechanism underlying IBR's predictive power deserves deeper theoretical and empirical exploration. While our results suggest that interest burden changes contain valuable information about future returns, the specific channels through which this information translates into stock price movements remain unclear.

Future research should pursue several promising directions. First, investigating the interaction between IBR and macroeconomic conditions, particularly interest rate cycles and credit market conditions, could provide insights into when the signal is most effective. Second, examining the signal's performance in international markets would test its generalizability beyond U.S. equities. Third, combining IBR with machine learning techniques or other non-linear modeling approaches might uncover additional predictive patterns. Finally, exploring the real economic outcomes associated with IBR changes—such as subsequent investment, profitability, and financial distress—would enhance our understanding of why this signal predicts returns. Despite these areas for future investigation, our findings establish IBR as a robust addition to the cross-sectional return prediction literature, offering both theoretical insights into market efficiency and practical value for portfolio construction.

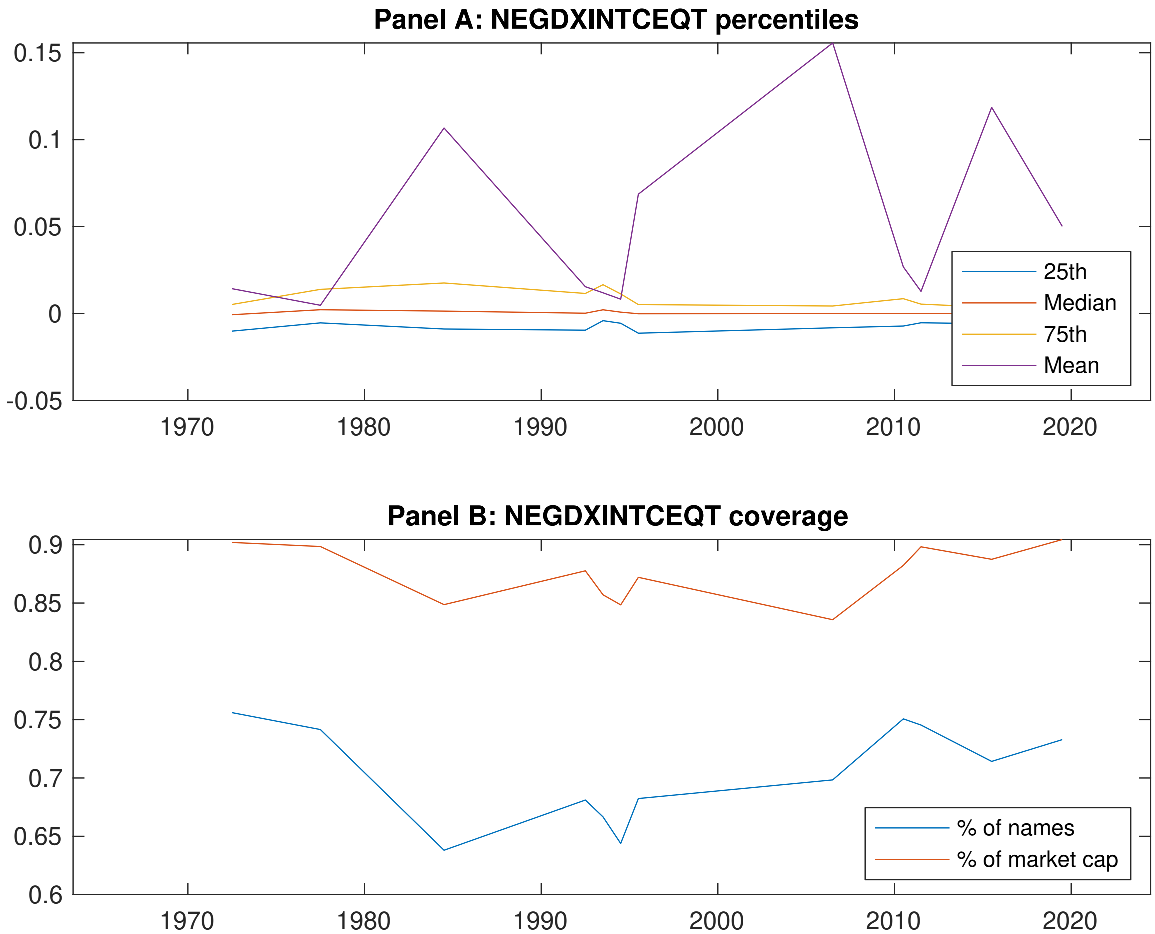


Figure 1: Times series of IBR percentiles and coverage.
This figure plots descriptive statistics for IBR. Panel A shows cross-sectional percentiles of IBR over the sample. Panel B plots the monthly coverage of IBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on IBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on IBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.52	0.64	0.56	0.60	0.69	0.16
	[2.75]	[4.00]	[3.51]	[3.66]	[3.86]	[2.36]
α_{CAPM}	-0.13	0.09	0.01	0.04	0.08	0.21
	[-2.54]	[2.07]	[0.28]	[0.85]	[1.53]	[3.05]
α_{FF3}	-0.16	0.08	0.01	0.08	0.07	0.22
	[-3.09]	[1.89]	[0.31]	[1.72]	[1.23]	[3.17]
α_{FF4}	-0.15	0.06	0.03	0.08	0.09	0.24
	[-2.84]	[1.36]	[0.76]	[1.82]	[1.64]	[3.30]
α_{FF5}	-0.17	-0.00	-0.00	0.08	-0.01	0.16
	[-3.37]	[-0.02]	[-0.06]	[1.65]	[-0.22]	[2.25]
α_{FF6}	-0.16	-0.01	0.02	0.08	0.02	0.18
	[-3.16]	[-0.29]	[0.39]	[1.76]	[0.28]	[2.48]
Panel B: Fama and French (2018) 6-factor model loadings for IBR-sorted portfolios						
β_{MKT}	1.10	0.99	0.96	0.96	1.03	-0.07
	[88.93]	[95.63]	[93.15]	[86.88]	[80.49]	[-4.01]
β_{SMB}	0.09	-0.06	-0.10	-0.08	0.13	0.04
	[5.04]	[-3.87]	[-6.45]	[-4.99]	[6.87]	[1.49]
β_{HML}	0.08	-0.00	-0.04	-0.11	-0.04	-0.13
	[3.47]	[-0.18]	[-2.10]	[-5.25]	[-1.81]	[-3.83]
β_{RMW}	0.09	0.16	-0.02	-0.02	0.19	0.09
	[3.91]	[7.85]	[-0.95]	[-0.98]	[7.48]	[2.75]
β_{CMA}	-0.08	0.13	0.13	0.05	0.11	0.18
	[-2.22]	[4.50]	[4.33]	[1.64]	[2.90]	[3.75]
β_{UMD}	-0.01	0.02	-0.03	-0.01	-0.04	-0.03
	[-0.97]	[1.63]	[-2.69]	[-0.79]	[-3.03]	[-1.55]
Panel C: Average number of firms (n) and market capitalization (me)						
n	607	501	651	577	612	
me ($\$10^6$)	1390	2200	2334	2024	1314	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the IBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.16	0.21	0.22	0.24	0.16	0.18
			[2.36]	[3.05]	[3.17]	[3.30]	[2.25]	[2.48]
Quintile	NYSE	EW	0.32	0.36	0.38	0.36	0.37	0.35
			[6.33]	[7.26]	[7.59]	[6.98]	[7.40]	[6.95]
Quintile	Name	VW	0.15	0.20	0.20	0.22	0.14	0.16
			[2.11]	[2.78]	[2.80]	[2.97]	[1.88]	[2.15]
Quintile	Cap	VW	0.15	0.19	0.21	0.20	0.14	0.13
			[2.33]	[3.11]	[3.29]	[3.07]	[2.15]	[2.10]
Decile	NYSE	VW	0.31	0.37	0.38	0.36	0.30	0.29
			[3.42]	[4.13]	[4.23]	[3.98]	[3.26]	[3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.12	0.17	0.18	0.19	0.13	0.14
			[1.69]	[2.45]	[2.56]	[2.68]	[1.78]	[1.94]
Quintile	NYSE	EW	0.10	0.15	0.16	0.15	0.13	0.13
			[1.68]	[2.47]	[2.59]	[2.45]	[2.17]	[2.10]
Quintile	Name	VW	0.10	0.15	0.15	0.16	0.10	0.11
			[1.43]	[2.14]	[2.14]	[2.29]	[1.38]	[1.56]
Quintile	Cap	VW	0.11	0.16	0.17	0.17	0.11	0.11
			[1.70]	[2.59]	[2.75]	[2.66]	[1.79]	[1.79]
Decile	NYSE	VW	0.25	0.32	0.33	0.32	0.25	0.25
			[2.78]	[3.57]	[3.63]	[3.53]	[2.81]	[2.79]

Table 3: Conditional sort on size and IBR

This table presents results for conditional double sorts on size and IBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on IBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high IBR and short stocks with low IBR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

	IBR Quintiles					IBR Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.48	0.86	0.95	0.96	0.69	0.22	0.24	0.24	0.23	0.20	0.20
	[1.75]	[3.49]	[3.60]	[4.08]	[2.67]	[2.54]	[2.78]	[2.74]	[2.60]	[2.25]	[2.20]
(2)	0.72	0.80	0.81	0.90	0.83	0.11	0.15	0.14	0.13	0.16	0.15
	[2.88]	[3.54]	[3.79]	[4.03]	[3.50]	[1.15]	[1.68]	[1.56]	[1.40]	[1.67]	[1.55]
(3)	0.62	0.82	0.73	0.76	0.93	0.31	0.35	0.37	0.34	0.37	0.34
	[2.65]	[4.20]	[3.70]	[3.76]	[4.25]	[3.67]	[4.16]	[4.40]	[3.95]	[4.27]	[3.94]
(4)	0.64	0.65	0.80	0.74	0.74	0.10	0.12	0.13	0.13	0.07	0.08
	[3.12]	[3.58]	[4.31]	[3.84]	[3.74]	[1.25]	[1.59]	[1.70]	[1.67]	[0.92]	[1.00]
(5)	0.47	0.61	0.49	0.59	0.61	0.14	0.19	0.19	0.18	0.11	0.11
	[2.62]	[3.89]	[3.09]	[3.51]	[3.63]	[1.70]	[2.39]	[2.36]	[2.23]	[1.40]	[1.40]

Panel B: Portfolio average number of firms and market capitalization

	IBR Quintiles: Average n					IBR Quintiles: Average market capitalization (\$$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	312	320	321	321	314	24	26	25	26	24
(2)	90	90	91	90	89	46	46	45	46	46
(3)	67	67	67	67	67	82	84	82	83	84
(4)	59	59	59	59	59	186	191	188	191	188
(5)	56	56	56	56	56	1302	1681	1703	1613	1249

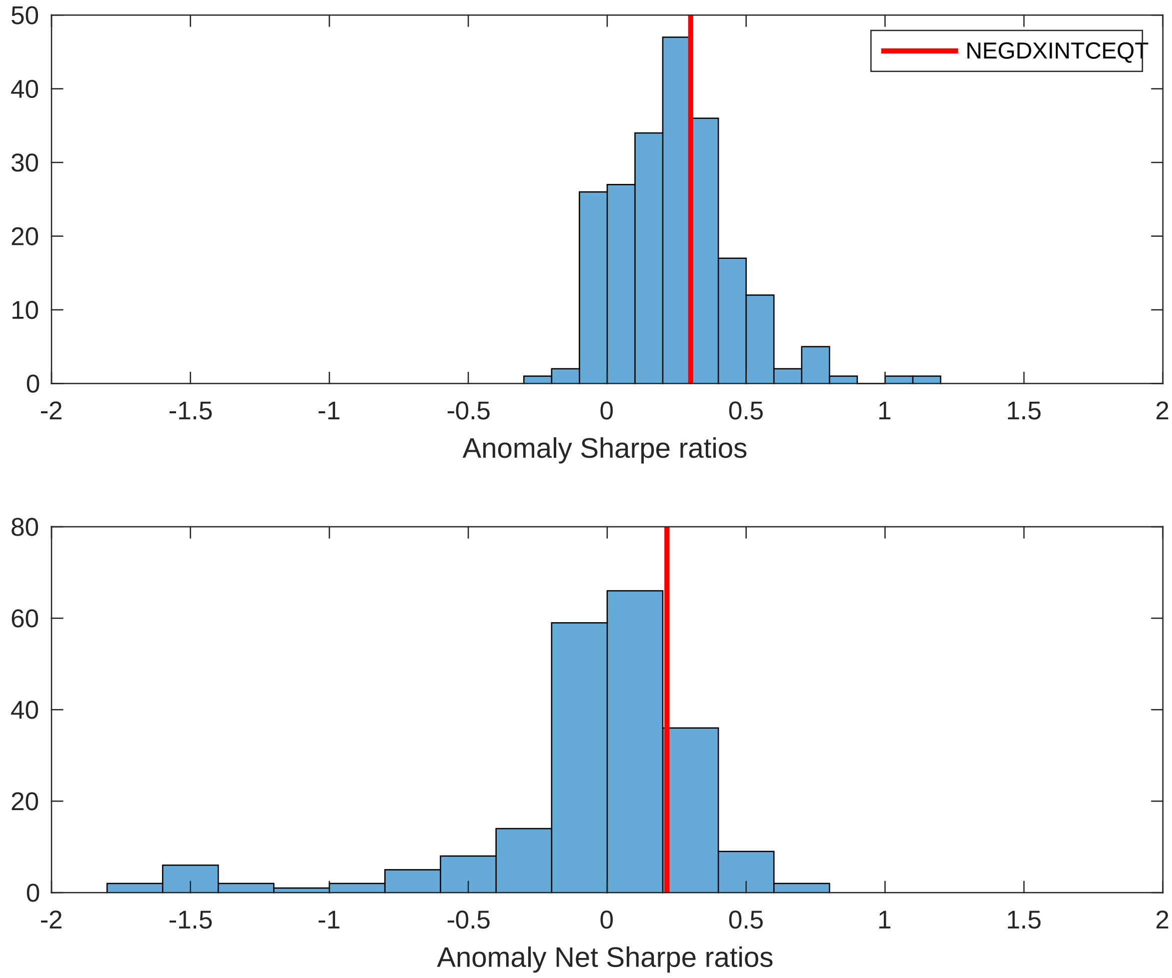


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the IBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

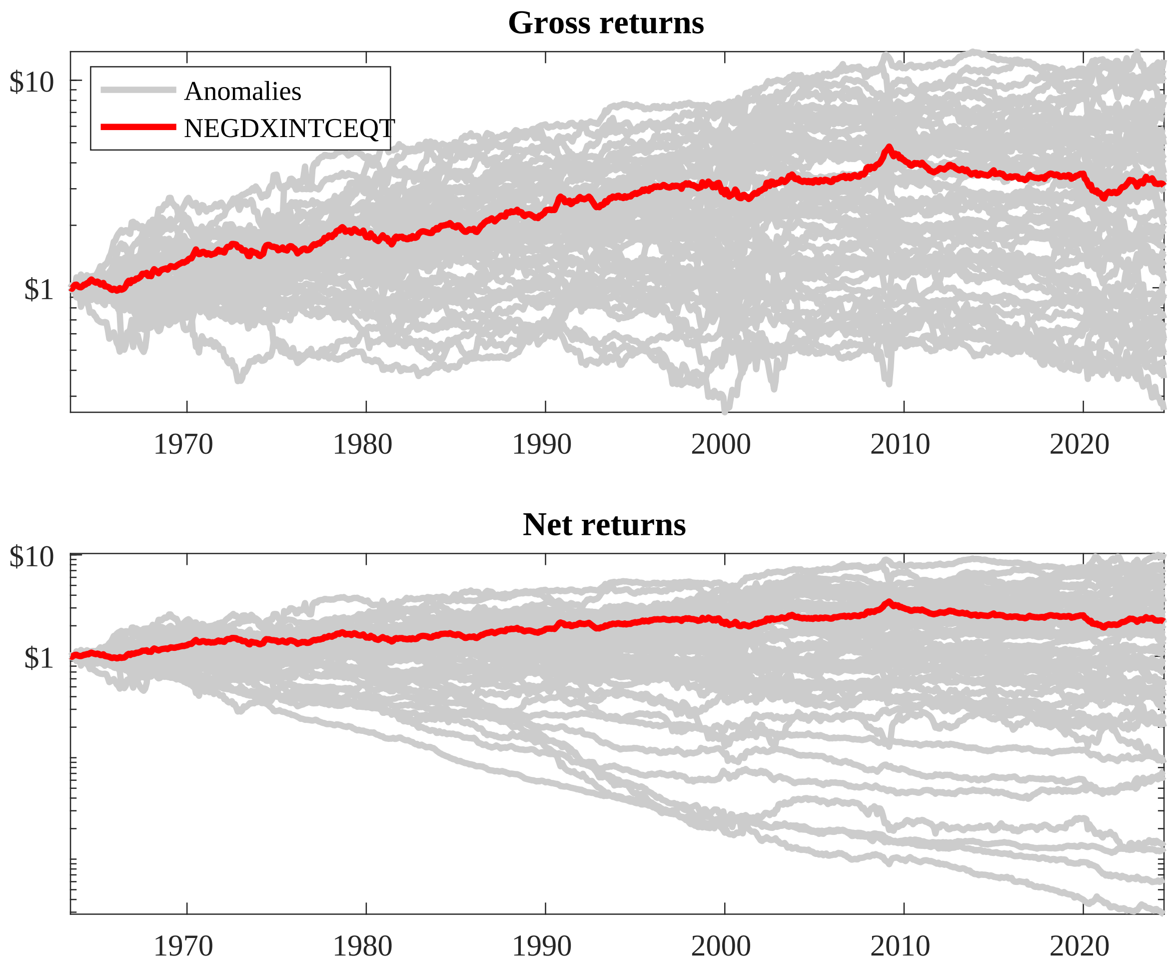
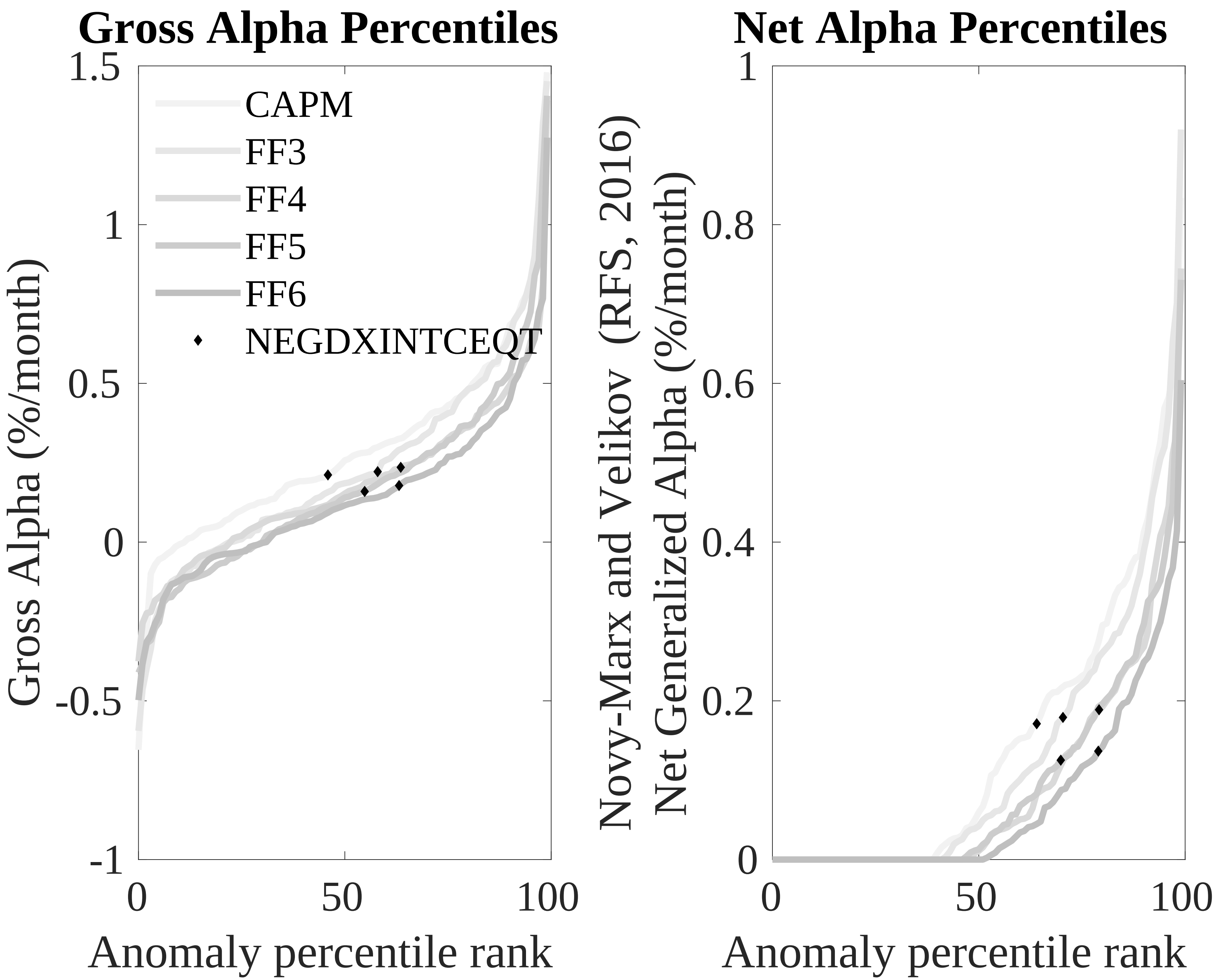


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the IBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.



Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDXINTCEQT
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100

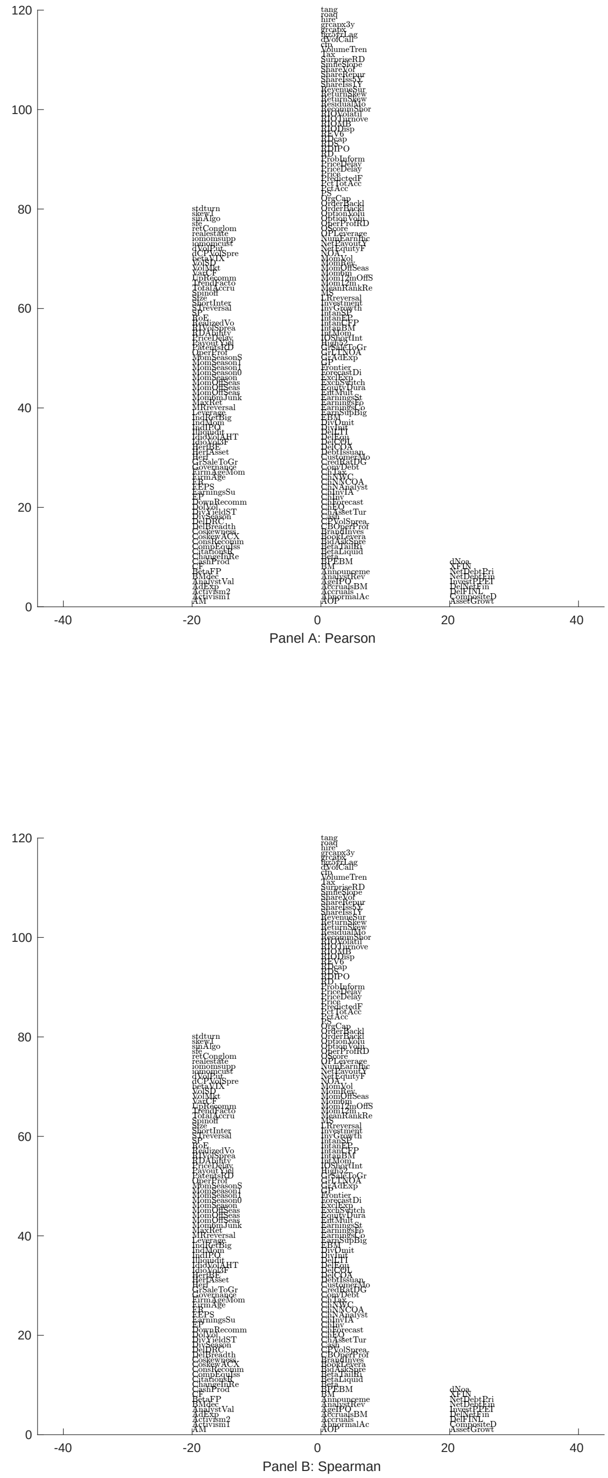


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 209 filtered anomaly signals with IBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

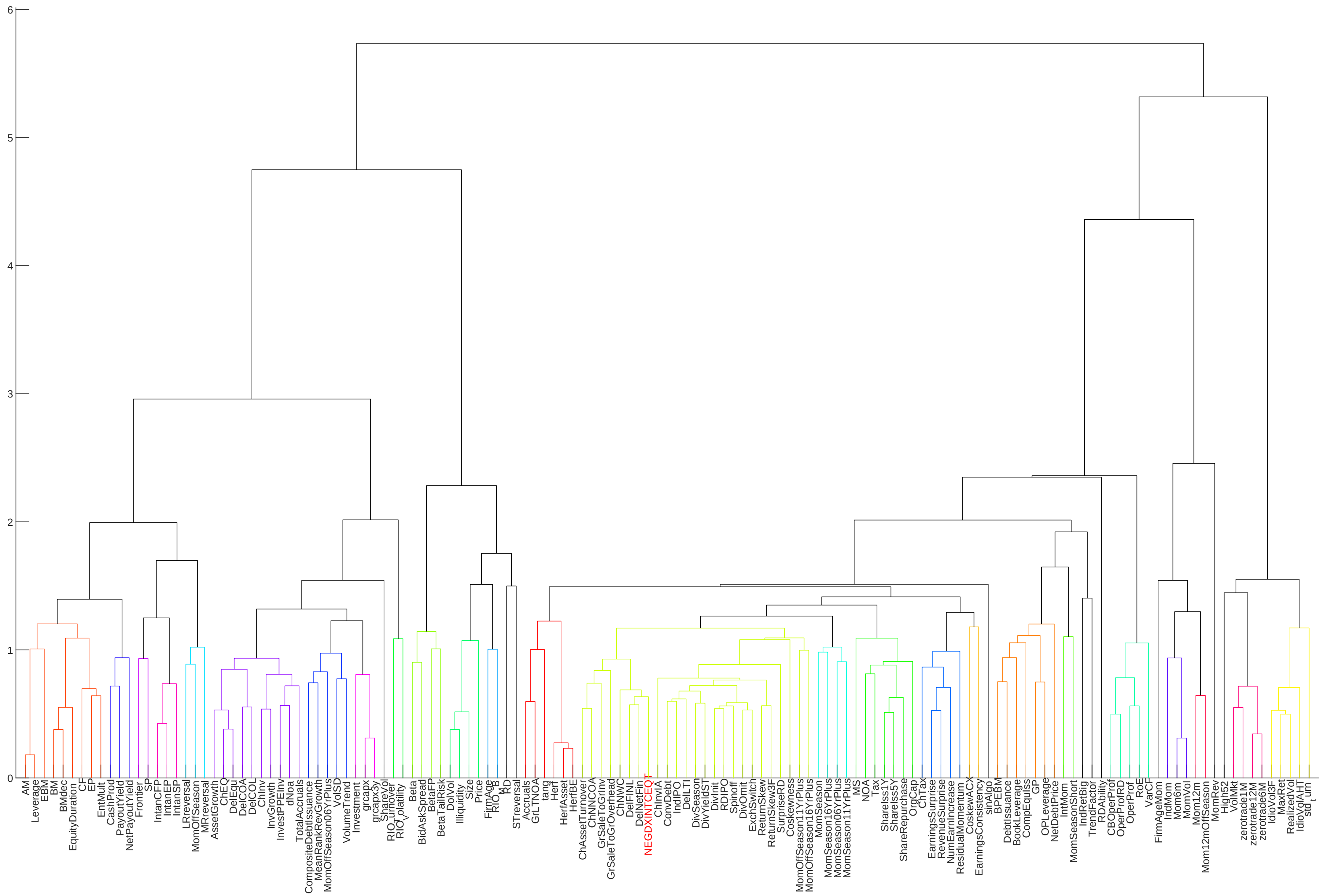


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

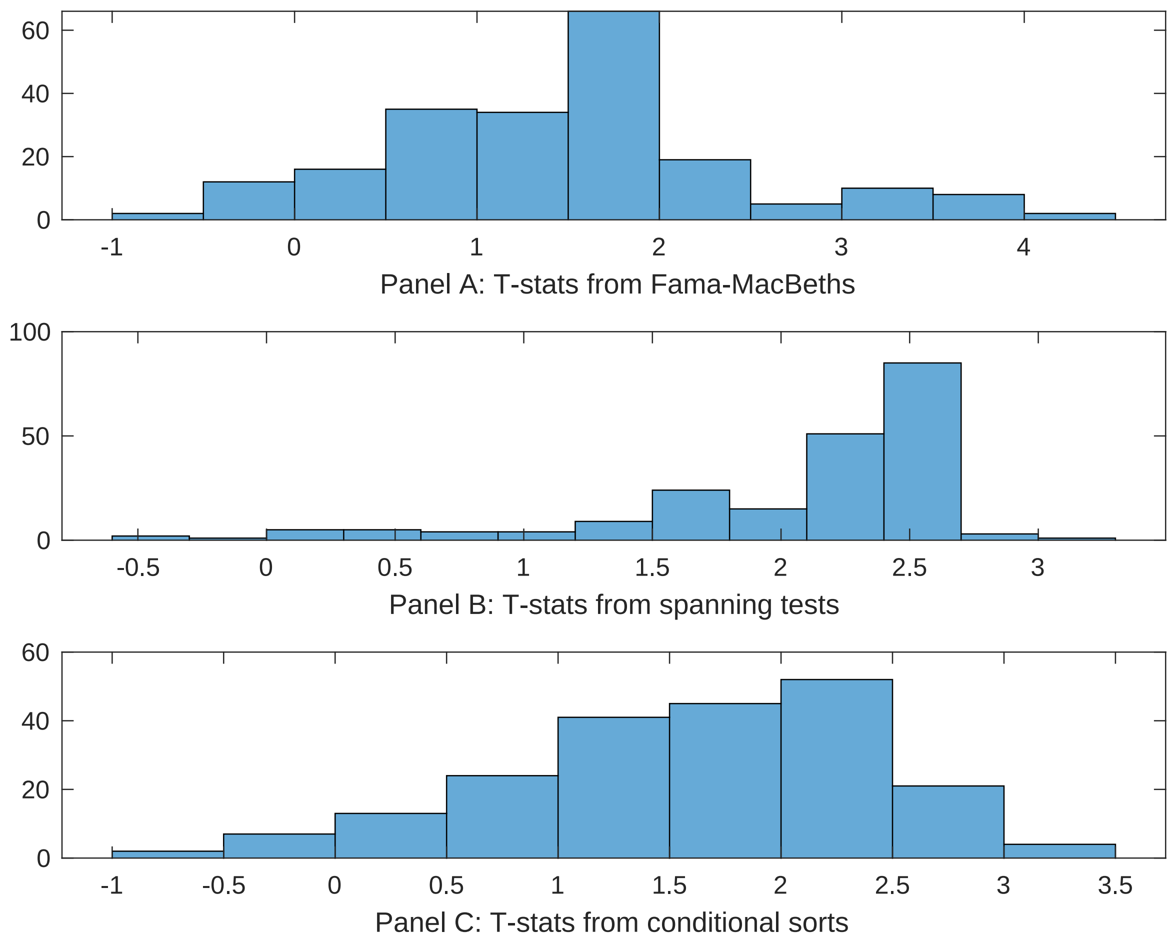


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of IBR conditioning on each of the 209 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{IBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{IBR} IBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 209 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{IBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 209 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 209 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on IBR. Stocks are finally grouped into five IBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted IBR trading strategies conditioned on each of the 209 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on IBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{IBR} IBR_{i,t} + \sum_{k=1}^{s} ix \beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.12	0.12	0.13	0.13	0.13	0.10	0.11
	[5.67]	[5.55]	[5.17]	[6.31]	[5.49]	[4.35]	[4.50]
IBR	-0.54	0.19	0.70	0.11	0.61	0.45	-0.31
	[-0.10]	[0.36]	[2.72]	[1.41]	[2.45]	[0.55]	[-0.10]
Anomaly 1	0.18						0.62
	[10.31]						[1.20]
Anomaly 2		0.96					0.30
		[6.77]					[0.09]
Anomaly 3			0.20				-0.54
			[9.25]				[-0.78]
Anomaly 4				0.95			0.56
				[5.05]			[3.18]
Anomaly 5					0.21		0.14
					[7.23]		[2.20]
Anomaly 6						0.20	0.13
						[2.69]	[2.55]
# months	732	732	630	732	630	732	628
$\bar{R}^2$(%)	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the IBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{IBR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in financial liabilities, Change in net financial assets, Net debt financing, Composite debt issuance, Net external financing, Revenue Growth Rank. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.15	0.14	0.13	0.19	0.13	0.24	0.15
	[2.20]	[1.95]	[1.72]	[2.71]	[1.64]	[3.44]	[1.99]
Anomaly 1	35.32						28.63
	[8.84]						[4.80]
Anomaly 2		23.07					13.59
		[6.81]					[3.40]
Anomaly 3			21.52				-10.76
			[5.03]				[-1.92]
Anomaly 4				15.94			5.15
				[4.71]			[1.38]
Anomaly 5					10.90		8.93
					[2.75]		[2.32]
Anomaly 6						17.54	16.50
						[5.93]	[5.15]
mkt	-6.12	-6.60	-5.88	-6.08	-4.36	-5.93	-2.40
	[-3.72]	[-3.94]	[-3.25]	[-3.55]	[-2.27]	[-3.50]	[-1.34]
smb	0.03	5.01	1.89	1.22	7.46	1.90	2.54
	[0.01]	[2.06]	[0.69]	[0.48]	[2.46]	[0.77]	[0.85]
hml	-9.93	-12.77	-12.04	-14.87	-11.96	-15.58	-13.54
	[-3.11]	[-3.94]	[-3.47]	[-4.48]	[-3.39]	[-4.73]	[-4.03]
rmw	5.95	12.00	7.68	7.56	3.67	5.56	0.75
	[1.85]	[3.64]	[2.16]	[2.27]	[0.86]	[1.66]	[0.19]
cma	4.43	21.93	11.29	13.73	10.21	4.45	-6.64
	[0.90]	[4.57]	[2.10]	[2.79]	[1.73]	[0.84]	[-1.08]
umd	-5.51	-3.17	-4.24	-3.43	-3.06	-3.08	-5.47
	[-3.32]	[-1.91]	[-2.34]	[-2.02]	[-1.68]	[-1.84]	[-3.19]
# months	732	732	630	732	630	732	630
$\bar{R}^2$(%)	14	11	8	8	6	10	20

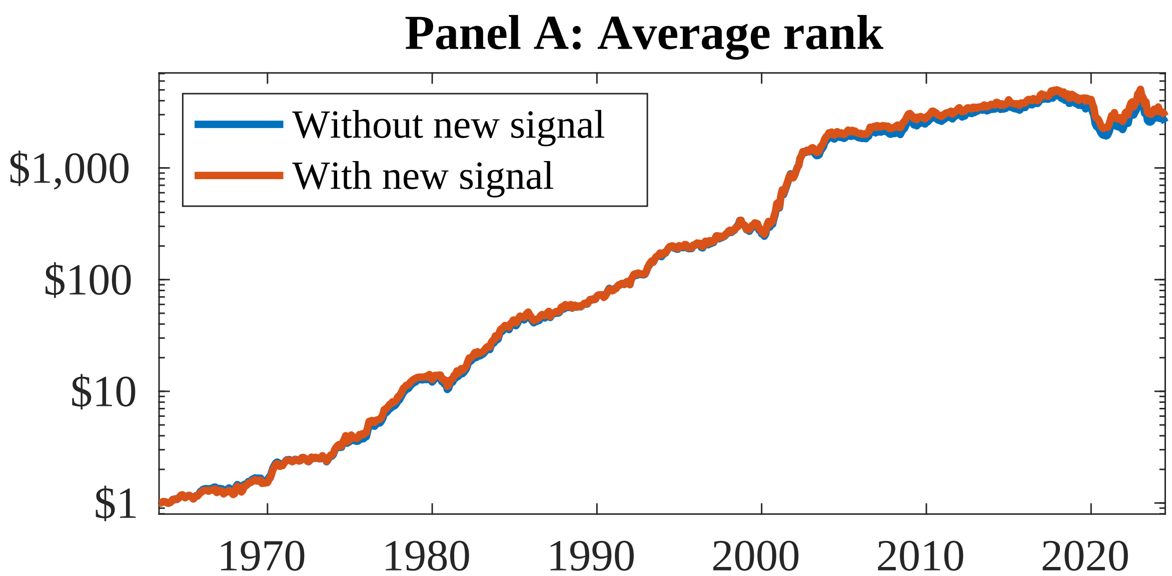


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as IBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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